Ambident Nucleophiles

Nucleophiles such as cyanide and nitrite ion which can attack nucleophilic centre from two sides of the nucleophile are called ambident nucleophiles. These nucleophiles can attack with either of the two sides depending upon the reaction conditions and the reagent used.

ii) Reaction with alcoholic KCN

Haloalkanes react with alcoholic KCN solution to form alkyl cyanides.

Example

$CH_3-CH_2-Br + KCN \rightarrow CH_3-CH_2-CN + KBr$

Bromoethane — Ethyl cyanide

iii) Reaction with alcoholic AgCN

Haloalkanes react with alcoholic AgCN solution to form alkyl isocyanide.

Example

$CH_3CH_2Br + AgCN \longrightarrow CH_3CH_2NC + AgBr$

Bromoethane — Ethyl isocyanide

iv) Reaction with sodium or potassium nitrite

Haloalkanes react with alcoholic solution of $NaNO_2$ or KNO_2 to form alkyl nitrites.

Example

$CH_3CH_2Br + KNO_2 \rightarrow CH_3CH_2-O-N=O + KBr$

Bromoethane — Ethyl nitrite

v) Reaction with silver nitrite

Haloalkanes react with alcoholic solution of $AgNO_2$ to form nitro alkanes.

Example

$CH_3CH_2Br + AgNO_2 \rightarrow CH_3CH_2NO_2 + AgBr$

Bromo ethane — Nitro ethane

vi) Reaction with sodium or potassium hydrogen sulphide

Haloalkanes react with sodium or potassium hydrogen sulphide to form thio alcohols.

Example

$$CH_3CH_2Br + NaSH \xrightarrow[\Delta]{\text{alcohol/ } H_2O} CH_3CH_2SH + NaBr$$

Bromo ethane — Ethane thiol

vii) Williamson ether synthesis

Haloalkane, when boiled with sodium alkoxide gives corresponding ethers.

Example

This method can be used to prepare mixed (unsymmetrical) ethers also.

$CH_3CH_2Br + NaOCH_2CH_3$

Bromo ethane — Sodium ethoxide

$\downarrow \triangleright$

$CH_3CH_2OCH_2CH_3 + NaBr$

diethyl ether

Mechanism of Nucleophilic substitution reaction

The mechanism of nucleophilic substitution reaction is classified as

a) Bimolecular Nucleophilic substitution reaction (SN_2)

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